

Yanrui Daisy Ning, Ph.D.

Assistant Professor, Environmental Engineering and Earth Science

Clemson University

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Research Profile

My research develops predictive multiphysics frameworks to understand how fluids, stresses, temperature, and reactions co-evolve in porous and fractured subsurface systems, enabling quantitative assessment of stability and leakage risks. I integrate coupled thermal-hydrological-mechanical-reactive (THM-R) modeling with distributed fiber-optic sensing (DFOS) and physics-informed machine learning to quantify multiphase transport, fracture propagation, and fluid-rock interaction. These approaches support risk-informed design of oil and gas production, carbon sequestration, hydrogen storage, geothermal energy, critical mineral systems, and hydrogeologic applications.

Education

Colorado School of Mines

Aug 2013 - Dec 2017

- Ph.D. in Petroleum Engineering (Major) & Geophysics (Minor), GPA: 3.95/4.0
- Dissertation: *Production Potential of Niobrara and Codell: Integrating Reservoir Simulation with 4D Seismic and Microseismic Interpretation*. Advisor: Hossein Kazemi

University of Wyoming, M.S. in Environmental Engineering

Aug 2011 - Aug 2013

China University of Petroleum, B.S. in Petroleum Engineering

Sep 2007 - Jul 2011

Grants

Funded Projects:

- PI, Michigan Tech Research Excellence Fund (REF), *Fiber-Optic Monitoring of Excavation- and Seepage-Induced Drift Deformation for Safer Critical-Mineral Mining*, **\$26,960**. Jan 2026 - Dec 2026.
- PI, DOE, *CarbonSAFE CO₂ Storage Project (DE-FE0032450)*, **\$11.1 million**. Aug 2024 - Oct 2025.
- Senior Personnel, DOE, *CCUS Pre-feasibility Study in Colorado (DE-FE0031837)*, **\$491K**. Oct 2021 - Sep 2024.
- Senior Personnel, DOE, *Carbon Utilization and Storage Partnership (CUSP) for Western US (DE-FE0031837)*, **\$840K**. Oct 2021 - Oct 2024.

Submitted Proposals:

- PI, DOE Office of Science (BES), *Quantifying the Degrees of Chemo-mechanical Coupling Governing Porosity and Permeability in Reactive Mineral Systems*, **\$867,822**. Submitted Nov 2025.

Professional Experience

Clemson University, Assistant Professor

May 2026 - Present

- Lead graduate researchers in developing coupled thermal-hydrological-mechanical simulation models to understand fiber optic sensing signals, with integration of AI-based analytical approaches.

Michigan Technological University, Assistant Professor

Aug 2025 - May 2026

- Supervise an MS student in developing flow-geomechanics simulation models for mineral extraction.
- Teach undergraduate courses in data analytics and geological MODFLOW modeling.
- Organize IMAGE Post-convention Workshop (PCW) on critical mineral and rare earth mineral extraction.

Colorado School of Mines, Research Assistant Professor/Research Associate/Postdoc

Jan 2021 - Aug 2025

- Lead multi-disciplinary teams to design CO₂ injection strategies, develop fluid flow and leakage-risk models, and coordinate stakeholder engagement for large-scale DOE-funded carbon storage projects.
- Co-lead field-scale geothermal testing at Utah, leveraging Distributed Acoustic Sensing (DAS) and Distributed Strain Sensing (DSS) technologies to optimize hydraulic fracture stimulation and cross-well connectivity; applying machine learning models to optimize the production well geometry.
- Developed coupled hydro-mechanical models and Python-based fiber-optic data workflows to monitor deformation and predict fracture behavior in subsurface energy systems.
- Identified fracture networks and optimized well placement in unconventional reservoirs by integrating stress fields, natural fracture data, and DAS data

- Developed production type curves for Washakie Basin by leveraging large-scale data analytics using Power BI, Python, and Spotfire; automated long-term data analysis, reducing the project time by 75% and saving \$20K/year in software license fees
- Increased well performance by 30% by optimizing well planning and forecasting production using decline curve analysis (DCA), analytical modeling, and reserves estimation (EUR)
- Designed optimized hydraulic fracture stage spacing and modeled stimulated reservoir volumes using heterogeneous fluid-flow simulations; conducted economic evaluations to identify the most profitable operating strategies.

Publications

Journal Publications ([Google Scholar](#))

Reservoir Simulation of Fluid Flow and Geomechanical Coupling:

1. **Ning, Y.**, Tura A., Herman, D., Bridgeman J., and Clark D., 2025. Feasibility Study on Large-Scale Geologic Carbon Sequestration in Southern Colorado. *International Journal of Greenhouse Gas Control*, 147, p.104462.
2. **Ning, Y.**, Tura A., and Guiltinan E., 2025. Repurposing a Depleted Natural CO₂ Dome for Permanent CO₂ Sequestration. *SPE Journal*, pp.1-15.
3. **Ning, Y.**, Boak J., Tura A., and Prasad M., 2024. Assessment of CO₂ Sequestration Potential and Economics in Colorado, USA. *International Journal of Greenhouse Gas Control*, 141, p.104301.
4. **Ning, Y.** and Tura, A., 2023. Economic and operational investigation of CO₂ sequestration through enhanced oil recovery in unconventional reservoirs in Colorado, USA. *Geoenergy Science and Engineering*, p.211820.
5. **Ning, Y.**, Kazemi, H., Tura, A. and Davis, T., 2023. Carbon neutral oil assessment using a dual-porosity compositional model in unconventional reservoirs. *Geoenergy Science and Engineering*, 221, p.111227.
6. **Ning, Y.**, Kazemi, H., Tura, A. and Davis, T.L., 2022. Tracer analysis in flow channel characterization and modelling of gas and CO₂ injection EOR in unconventional reservoirs. *Journal of Petroleum Science and Engineering*, 212, p.110349.
7. **Ning Y.**, Wang L., Yu X., and Li J., 2023. Recent development in the decarbonization of marine and offshore engineering systems. *Ocean Engineering*, 280, p.114883.
8. **Ning, Y.**, Jin, G., Hammack, R., Zorn, E., Kohnke, C., and Rosenbaum, E. 2025. Monitor Potential Fracture Growth in Carbon Sequestration Reservoirs Using Distributed Strain Sensing. Minor revision in process. *SPE Journal*.

Distributed Fiber Optic Sensing (DFOS) & Subsurface Fracture Characterization:

9. **Ning, Y.**, Jin, G., Tura A., and Gale M., 2024. Exploring Variations in Orientation and Velocity of Hydraulic Fracture Propagation in Denver-Julesburg Basin. *SPE Journal*, pp.1-12.
10. Jin, G., **Ning, Y.**, Tura A., and Gale M., 2024. Impact of natural fractures on hydraulic fracture propagation in Denver-Julesburg Basin: Insights from a decade of research. *The Leading Edge*, 43(12), pp.806-814.
11. **Ning, Y.** and Jin G., 2023. Challenges and best practices in interpreting cross-well strain signals to monitor multi-crew zipper fracturing operations. *Interpretation*. 11(2), pp.1-43.
12. **Ning, Y.**, Jin, G., Zhu, X. and Tura, A., 2022. Well interference and fracture geometry investigation using production and Low-Frequency Distributed Acoustic Sensing data in an unconventional reservoir. *SPE Reservoir Evaluation & Engineering*, pp.1-11.
13. **Ning Y.**, Jin G., Hammack R., Kohnke C., Rosenbaum E., and Zorn E., 2025. Subsurface Fracture Characterization via Inversion of Distributed Strain Sensing Data. *SPE Journal*. Revision submitted.

Machine Learning for Subsurface Energy Production:

14. **Ning, Y.**, Schumann H., and Jin G. 2022. Application of data mining to small datasets: identification of key production drivers in unconventional resources. *SPE Reservoir Evaluation & Engineering*. SPE-212301-PA. pp.1-11.
15. **Ning, Y.**, Kazemi, H. and Tahmasebi, P., 2022. A comparative machine learning study for time series oil production forecasting: ARIMA, LSTM, and Prophet. *Computers & Geosciences*, 164, p.105126.

16. **Ning, Y.**, Bailey, J.R., Bourdier, J., Prasad, P. and Momoh, I., 2024. Optimizing Production Well Geometry in the Utah FORGE Geothermal Project Using Machine Learning and Fluid Flow Modeling. *Renewable Energy*, p.121767.

Hydrogen Subsurface Storage:

17. **Ning Y.**, Wang L., Wang R., Wang H., and Li J. Enhancing Hydrogen Storage Recovery in the Sundance Formation: Operational Optimization in the Powder River Basin, USA. *Green Energy & Environment*, Submitted.
18. Wang, H., Xin, Y., Kou, Z., Qu, Y., Wang, L., **Ning, Y.** and Ren, D., 2023. Numerical study of the efficiency of underground hydrogen storage in deep saline aquifers, rock springs uplift, Wyoming. *Journal of Cleaner Production*, 421, p.138484.

Peer-Reviewed Book Chapters

19. **Ning, Y.** and Jin G. Enhancing unconventional reservoir development through low-frequency distributed acoustic sensing technology. *Distributed Acoustic Sensing in Borehole Geophysics*, pp.499-530.

Conference Presentations (Past 5 years)

1. **Ning, Y.** and , , Presented at the *IMAGE* 2026 Conference, Houston, TX.
2. **Ning, Y.**, Jin, G., Hammack, R., Zorn, E., Kohnke, C., and Rosenbaum, E. 2025. Monitor potential fracture growth in carbon sequestration reservoirs using distributed strain sensing. Presented at the *IMAGE* 2025 Conference, Houston, TX.
3. Bailey, J.R., **Ning, Y.**, Bourdier, J., Momoh, I. and Prasad, P., 2024, January. Application of Machine Learning to Create a Discrete Fracture Network Model for Utah FORGE Fracture Injections. In *SPE Hydraulic Fracturing Technology Conference and Exhibition* (p. D021S004R005). SPE.
4. Jin, G., **Ning, Y.**, Li, P., Hammack, R. and Wu, K., 2024. *Distributed Strain Sensing based Reservoir Monitoring for Carbon Sequestration Reservoirs*. National Energy Technology Laboratory (NETL), Pittsburgh, PA, Morgantown, WV, and Albany, OR (United States).
5. **Ning Y.** and Tura A., 2024. CO₂ Sequestration into A Depleted Natural CO₂ Dome: Sheep Mountain, Colorado. In *International Meeting for Applied Geoscience & Energy (IMAGE)* 2024.
6. **Ning Y.** and Jin G., 2024. Investigation of Hydraulic Fracture Orientation and Propagation Velocity in DJ Basin LF-DAS. Presented at the *IMAGE* 2024 Conference, Houston, TX.
7. **Ning Y.** and Tura A., 2024. Advancing Large-scale Geological Carbon Sequestration in Colorado. Presented at the *IMAGE* 2024 Conference, Houston, TX.
8. **Ning Y.** and Tura A., 2023. Coupled fluid-geomechanical modeling for a large-scale CO₂ sequestration project in Southern Colorado. In *International Meeting for Applied Geoscience & Energy* 2023.
9. **Ning Y.** and Jin G., 2023. Investigation on the azimuth variation and propagation of hydraulic fractures using low-frequency distributed acoustic sensing (LF-DAS). In *International Meeting for Applied Geoscience & Energy* 2023.
10. **Ning, Y.** and Tura, A. 2023. Technical Evaluation of CO₂ Capture from an NGCC Power Plant and Sequestration in Saline Aquifers (Colorado). In *Carbon Capture, Utilization, and Storage (CCUS) Conference*.
11. **Ning Y.** and Tura, A., 2022, September. Wellbore Integrity Impact on Carbon Leakage to Ensure Safe Geological Sequestration. In *SPE Annual Technical Conference and Exhibition*. OnePetro.
12. **Ning, Y.**, Mjehovich, J., Zhu, X. and Jin, G., 2022, November. Low-frequency DAS interpretation upon multi-crew zipper fracturing configuration. In *SEG International Exposition and Annual Meeting* (p. D011S031R001). SEG.
13. **Ning, Y.** and Tura, A. 2022. CO₂ Injection into Unconventional Reservoirs and Saline Aquifers, and Leakage Risk Assessment. In *AAPG Carbon Capture, Utilization, and Storage (CCUS) Conference*.
14. **Ning, Y.**, Jin, G., Zhu, X. and Tura, A., 2022, June. Integration of Production Communication and Hydraulic Fracture Connection in Unconventional Reservoirs. In *SPE/AAPG/SEG Unconventional Resources Technology Conference*. OnePetro.
15. Tura, A., **Ning, Y.** and Haj Nasser, Y., 2021, September. Climate change, the energy industry, and the role of carbon capture utilization and storage. In *First International Meeting for Applied Geoscience & Energy* (pp. 3004-3008). Society of Exploration Geophysicists.

Invited Presentations (Past 5 years)

1. University of Clemson, Department Environmental Engineering & Earth Science (Clemson, SC) Mar 2025

- Integrating Subsurface Modeling and Monitoring for Sustainable Geoenergy Futures
- 2. Michigan Tech University, Department of Geological and Mining Engineering and Sciences (Houghton, MI) Feb 2025
 - Advancing Subsurface Modeling and Monitoring for Geological Carbon Sequestration
- 3. Geophysical Research for Gigatonnes CO₂ Storage Workshop (Golden, CO) Jul 2024
 - CO₂ Sequestration into a Depleted Natural CO₂ Dome: Sheep Mountain, Colorado
- 4. University of Houston, Department of Earth and Atmospheric Sciences (Houston, TX) Mar 2024
 - Enhancing Energy Sustainability via Integrated Subsurface Characterization
- 5. University of Oklahoma, School of Geoscience (Norman, OK) Mar 2024
 - Advancing Large-Scale Geological Carbon Sequestration in Colorado
- 6. Georgia Institute of Technology, Geophysics Seminar Fall 2023 (Atlanta, GA) Nov 2023
 - Advancing Large-Scale Geological Carbon Sequestration in Colorado
- 7. Mines Geophysics 100th Anniversary, GP Gives - CCUS Workshop (Golden, CO) Nov 2023
 - Geological Carbon Sequestration Capacity Estimation in Colorado
- 8. 2023 Carbon Utilization Storage Partnership (CUSP) annual meeting (Lawrence, KS) Jun 2023
 - Feasibility Study of CO₂ Sequestration in Saline Aquifers in Southern Colorado
- 9. 2022 Carbon Management Project Review Meeting (Pittsburgh, PA) Aug 2022
 - Investigation of CO₂ Leakage to Drinking Water Formations During Carbon Sequestration
- 10. Toward Gigatonnes CO₂ Storage - Grand Geophysical Challenge Workshop (Stanford, CA) Jun 2022
 - CO₂ Leakage Risk Assessment in the Denver-Julesburg Basin, Colorado
- 11. 2022 Carbon Utilization Storage Partnership (CUSP) annual meeting (Las Vegas, NV) Jun 2022
 - Feasibility Study of a Potential CCUS Project in Colorado
 - CCUS progress update in Colorado
- 12. 27th Annual 3D Seismic Symposium (Denver, CO) Mar 2021
 - Integration of Production Communication and Hydraulic Fracture Connection in Unconventional Reservoirs

Teaching Experience

Courses as Primary Instructor

1. GE 3850 Geohydrology, Spring 2026
2. GEOL 4090 / 6090 Geophysics, Fall 2027

Courses with Shared Instruction

3. GE 1200 Introduction to Data Science, Fall 2025
 - Developed modules on data visualization
4. GE 1100 Geo-Orientation, Fall 2025
 - Share research with undergraduates
5. CCUS 521 Geological Carbon Capture Utilization and Sequestration, Fall 2023
 - Delivered case studies on coupled fluid flow-geomechanics modeling

Services

Co-organizer of

- IMAGE 2026 Post-conventional workshop - W8 Critical Minerals & Rare Earth Minerals (REE)
- SEG 2025 AI and its application in CCUS, EOR and the Energy Transition at Penn State University
- SEG 2024 Geologic Carbon Storage (GCS) Summer Research Workshop at Colorado School of Mines

Committee Member & Theme Chair & Session Chair of

- SPE/AAPG/SEG Carbon Capture, Utilization, and Storage (CCUS), 2024 - present
- SPE ATCE (Annual Technical Conference and Exhibition), 2023-2026
- IMAGE (The International Meeting for Applied Geoscience & Energy), 2023 - present
- OTC (SEG committee), 2021-2022

Reviewer for

- SPE Journal, Geophysics, Sustainability, Geoscience and Remote Sensing Letters, Geophysical Prospecting, Geoenergy Science and Engineering, Journal of Petroleum Science and Engineering,

International Journal of Greenhouse Gas Control, Petroleum Exploration and Production Technology, Petroleum Science and Technology, Energies, etc.